

Anterior Cruciate Ligament Reconstruction

A Prospective Randomized Study of Three Surgical Methods

Allen F. Anderson,* MD, Robert B. Snyder, MD, and A. Brant Lipscomb, Jr., MD

From The Lipscomb Clinic Sports Medicine Center and St. Thomas Hospital, Nashville, Tennessee

ABSTRACT

A prospective randomized study was performed to determine the differences in results between three methods of anterior cruciate ligament reconstruction: autogenous bone-patellar tendon-bone graft (group 1), semitendinosus and gracilis tendon graft reconstruction combined with an extraarticular procedure (group 2), and semitendinosus and gracilis tendon graft reconstruction alone (group 3). Preoperatively, there were no significant differences between groups. At a mean of 35.4 ± 11.6 months postoperatively, 102 patients returned for evaluation. International Knee Documentation Committee knee evaluation revealed no significant differences in symptoms, function, return to pre-injury activity, harvest site abnormalities, or limitation of motion between groups 1 and 3. Patients in group 2 had a higher incidence of patellofemoral crepitation and loss of motion than did patients in group 3. The mean manual maximum KT-1000 arthrometer side-to-side difference was 2.1 ± 2.0 mm in group 1, which was statistically significantly better than the difference in group 3 (3.1 ± 2.3 mm). Final knee rating showed that 34 of 35 patients in group 1, 23 of 34 patients in group 2, and 26 of 33 patients in group 3 had a normal or nearly normal overall knee rating. Anterior cruciate ligament reconstruction with a semitendinosus and gracilis or a patellar tendon autograft may yield similar subjective results; however, the patellar tendon autograft may provide better objective stability in the long term. In addition, there appears to be no benefit to combining an intraarticular anterior cruciate ligament reconstruction with an extraarticular procedure.

A variety of graft sources, such as autografts, allografts, and synthetic grafts, have been used for ACL reconstructions to improve function and kinematics in ACL-deficient knees. Despite the abundance of articles published about ACL reconstruction, there is not a consensus on which procedure or graft source is optimal. Presently, the autologous patellar tendon^{1-3, 7, 9, 10, 14, 15, 17, 25, 27} or semitendinosus and gracilis tendons^{6, 8, 11, 16, 18-20, 23, 28, 32-34, 36} are the most frequently used grafts for intraarticular ACL reconstruction.

The results of previous studies comparing the patellar tendon graft and the semitendinosus and gracilis tendon graft have been mixed. Harter et al.¹² did a retrospective analysis comparing 30 patients who had ACL reconstruction with patellar tendon and extraarticular augmentation and 20 patients who had reconstruction with semitendinosus and gracilis tendons and extraarticular augmentation. The authors found no statistically significant differences between these two methods. Holmes et al.¹³ performed a similar retrospective analysis and found that the patellar tendon ACL reconstruction was a statistically significantly better method for patients with chronic ACL insufficiency. Otero and Hutcheson²⁵ also found that patients who had patellar tendon reconstructions had greater stability and higher success rates when compared with patients undergoing doubled semitendinosus and gracilis tendon graft reconstructions.

In contrast, Marder et al.²¹ and Aglietti et al.² performed prospective comparative studies alternating the sequence of these two procedures. Neither of these studies demonstrated a statistically significant difference in the outcome, although there was a trend toward better stability in the group treated with patellar tendon reconstruction.²

Extraarticular reconstructions have been combined with both patellar tendon^{9, 24, 29-31} and semitendinosus and gracilis tendon intraarticular reconstructions^{18-20, 35} to serve as an adjunct that limits pathologic motion and protects the intraarticular graft. Roth et al.³⁰ reported no significant differences between patients who had an autologous patellar tendon ACL reconstruction either with

* Address correspondence and reprint requests to Allen F. Anderson, MD, Tennessee Orthopaedic Alliance, The Lipscomb Clinic, 4230 Harding Road, #1000, Nashville, TN 37205.

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or without an extraarticular biceps tendon advancement. Strum et al.³⁴ also found no difference between a group of 84 patients who had an intraarticular reconstruction with either a meniscus or patellar tendon and a group of 43 patients who had an extraarticular procedure combined with the intraarticular procedure. Noyes and Barber²³ compared 64 patients treated with an intraarticular patellar tendon allograft and 40 patients who had an intraarticular allograft and an extraarticular iliotibial band tenodesis. They found that the rate of success was improved by the addition of an extraarticular procedure. The authors concluded that the combination of intraarticular and extraarticular procedures is of value in young, athletically active people who have chronic ACL deficiency.

The purpose of this study was to determine prospectively the objective and subjective results of three types of ACL reconstruction procedures: 1) intraarticular reconstruction with bone-patellar tendon-bone autograft, 2) intraarticular reconstruction with semitendinosus and gracilis tendon combined with a Losee extraarticular iliotibial band tenodesis, and 3) intraarticular reconstruction with semitendinosus and gracilis tendons without an extraarticular procedure. To our knowledge, this is the first prospective, randomized study comparing these procedures.

MATERIALS AND METHODS

In 1991, a prospective, randomized study was designed to compare three methods of ACL reconstruction. The criteria for inclusion in the study were 1) willingness of the patient to be subjected to randomization, 2) no prior knee surgery other than meniscectomy, 3) a normal contralateral knee, 4) no demonstrative osteoarthritis of either knee, 5) no history of patellofemoral pain or significant patellofemoral crepitation, 6) age less than 50 years, and 7) absence of significant injury to other ligament structures, that is, less than a grade I injury to the medial collateral ligament.

From 1991 to 1993, a total of 267 patients with unilateral ACL tears were evaluated. The differences in the surgical procedures, risk of surgery, and potential benefits were carefully explained. Patients were asked to participate in a randomized study. Ninety-three patients were excluded because they did not want to receive randomized treatment, 32 were excluded for having an associated ligament injury, and 37 were excluded for failing to meet various other inclusion criteria.

A randomized list of procedures was generated by computer, and patients who met the inclusion criteria were assigned to one of three groups. Each patient was informed that he or she was being randomly treated with one of three surgical procedures, and all patients signed a voluntary consent form before inclusion in the study.

The 105 patients who were included in this study were assigned to 3 groups of 35. Group 1 consisted of patients who had intraarticular ACL reconstruction using an autologous bone-patellar tendon-bone graft. Group 2 patients had intraarticular ACL reconstructions with semitendinosus and gracilis tendon autografts combined with a Losee extraarticular iliotibial band tenodesis. Group 3

patients had intraarticular ACL reconstructions with semitendinosus and gracilis tendon autografts without an extraarticular procedure.

Most of the patients sustained injuries while participating in sports activities, with basketball ($N = 46$), football ($N = 25$), baseball ($N = 7$), skiing ($N = 5$), and soccer ($N = 2$) being the leading causes. Six patients were injured in other sports and 14 were injured in accidents.

Preoperative and postoperative evaluations included a comprehensive physical examination, KT-1000 arthrometer ligament laxity testing (MEDmetric Corp., San Diego, California), quadriceps and hamstring muscle strength testing (Cybex II, Cybex, International Inc., Medway, Massachusetts), and AP, lateral, and standing roentgenograms. Patients with chronic injuries were evaluated according to the criteria of the Hospital for Special Surgery knee score.³⁶ All the physical examinations and objective evaluations were performed by the senior author (AFA). In addition, patients were evaluated postoperatively using the criteria of the International Knee Documentation Committee (IKDC) Knee Evaluation Form.⁴ This method of evaluation was not included in the preoperative protocol because the form was not published until after the study began.

There were no statistically significant differences among the three groups preoperatively, including the variables of sex, age, length of time from injury to surgery, number of reinjuries, previous operations, preoperative activity level, symptoms, function, KT-1000 arthrometer laxity measurements, and roentgenographic changes (Table 1).

The incidence of a positive pivot shift preoperatively was relatively low (range, 46% to 57%). This finding may be explained by a high incidence of acute (<2 weeks) and subacute (2 to 12 weeks) injuries in each group. Anderson et al.⁵ previously documented that accuracy of the McIntosh pivot shift test was decreased if the length of time from injury to examination was less than 3 months.

Data Analysis

The patients were coded by number to preserve confidentiality and their data were entered into a series of linked databases. The data were analyzed using the SPSS statistical package (SPSS, Inc., Chicago, Illinois) on a personal computer. Scaled scores were calculated using a computer program.

All variables were examined by using univariate analysis to assess the data distribution and to help eliminate coding errors. Two-tailed statistical tests were used throughout. Differences between groups were assessed using contingency table analysis, analysis of variance with the Scheffé test for multiple comparisons or Bonferroni-adjusted *t*-tests and Kruskal-Wallis tests as applicable. The level of significance was set at $P \leq 0.05$.

The power was adjusted for an overall alpha of 0.05. The difference was measured in units of the standard deviation. For continuous data such as age, length, or continuous scores, a power of 0.7 detected a difference of 0.7 standard deviations. A power of 0.8 detected 0.8 standard

TABLE 1
Preoperative Comparison of the Three Groups

Category	Group ^a		
	1	2	3
Men (no.)	23	22	23
Women (no.)	12	13	12
Mean age (years)	23.6	22	20.1
Age range (years)	14–44	14–40	14–38
Time from injury to surgery (no. of patients)			
<2 weeks	6	13	9
2 to 12 weeks	17	10	18
>12 weeks	12	12	8
Previous arthroscopy (no. of patients)	6	3	4
Patellofemoral crepitus (no. of patients)			
1+	3	3	2
2+	0	0	1
3+	0	0	0
KT-1000 arthrometer manual maximum difference (mean $\pm$ SD)	5.3 $\pm$ 3.8	5.7 $\pm$ 2.8	6.0 $\pm$ 3.3
Pivot shift (no. of patients)			
0	17	19	15
1+	3	4	5
2+	8	6	10
3+	7	6	5

^a Group 1, surgery with an autogenous patellar tendon graft; group 2, surgery with a semitendinosus and gracilis graft and an extraarticular procedure; group 3, surgery with a semitendinosus and gracilis graft.

deviations. So if the difference between two groups was greater than 0.8 standard deviations, the power was greater than 0.8. For proportional or nominal data, such as percentage of people who can do something, the power was 0.7 to detect a 33% difference and 0.8 to detect a 36% difference.

Treatment Groups and Surgical Techniques

Group 1—Patellar Tendon Graft. The patients in group 1 had ACL reconstructions with the middle third of the patellar tendon. Thirteen patients in this group had a torn medial meniscus and 14 had a torn lateral meniscus. Three medial and 3 lateral menisci were repaired; 10 medial and 11 lateral menisci were partially excised.

The ACL reconstructions were performed using a single-incision endoscopic technique. An incision was made over the patellar tendon and the peritenon was split. The middle third of the patellar tendon (9 to 12 mm in diameter) was harvested with 25 to 30 mm of bone from the patella and the tibial tubercle. The bone was rounded to fit snugly through the same diameter sizer. A drill guide (Acufex Microsurgical, Inc., Norwood, Massachusetts) was used to drill a hole in the anteromedial aspect of the tibia, entering the joint in the posterior portion of the ACL footprint, across from the posterior edge of the anterior horn of the lateral meniscus. The soft tissue around the drill hole was carefully removed.

An Acufex isometer was used to identify the most isometric point on the lateral femoral condyle. A guide wire was inserted through the tibial hole into this location and it was over-reamed with the appropriate sized reamer. The edges of the hole were chamfered and the debris was removed throughout the joint. The tibial bone plug was

pushed through the tibial hole into the femoral hole and secured with a 7 $\times$ 25 mm interference screw. The graft was externally rotated 90°. The knee was extended to determine whether there was graft impingement on the roof of the intercondylar notch; if there was, additional notchplasty was performed.

With the knee flexed to 10°, the graft was placed under approximately 15 pounds of tension and secured in a tibial trough with two barbed staples (Fig. 1). Laxity was evaluated manually. The defects in the patellar tendon, the paratenon, subcutaneous tissue, and skin were closed in a routine manner.

Group 2—Semitendinosus and Gracilis Tendons with Extraarticular Procedure. The patients in group 2 had ACL reconstructions with the semitendinosus and gracilis tendon autograft combined with a Losee extraarticular iliotibial band tenodesis. Fourteen patients in group 2 had a torn medial meniscus and 15 had a torn lateral meniscus. Seven medial menisci and 1 lateral meniscus were repaired, and 7 medial and 14 lateral menisci were partially excised.

A 5-cm transverse incision was made over the pes anserinus tendon. The gracilis and semitendinosus tendons were dissected free in the distal thigh, divided at the musculotendinous junction, and sutured together. A whipstitch of 1–0 nonabsorbable material was placed in the tendon ends. A 12- to 15-cm incision was made over the lateral aspect of the knee, ending at Gerdy's tubercle. A strip of the iliotibial band, 2.5-cm wide and 15- to 20-cm long, was mobilized from proximal to distal, leaving it attached to Gerdy's tubercle.

Arthroscopic surgery was performed to treat meniscal tears and to perform the intraarticular ACL reconstruction. The debris was removed from the intercondylar

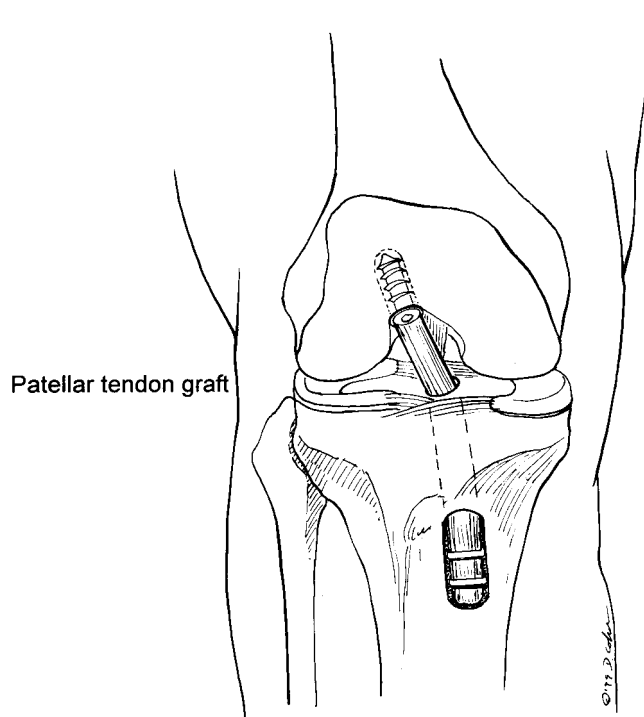


Figure 1. Patients in group 1 had ACL reconstruction with the middle third of the patellar tendon.

notch and a minimal notchplasty was performed. The tendon grafts were sized (6 to 8 mm). The Acufex drill guide was used to assist in drilling a hole in the antero-medial aspect of the tibia, entering the joint at the anatomic insertion of the ACL. The isometer was then used to identify the most isometric position on the lateral femoral condyle. The guide wire was inserted at this point and was over-reamed with the appropriate sized drill. The edges of the holes were chamfered.

Next, a 7- to 10-mm drill was used to make an anterior-to-posterior hole in the lateral femoral condyle beginning just anterior and superior to the femoral origin of the lateral collateral ligament, exiting at the origin of the lateral head of the gastrocnemius muscle.

The semitendinosus and gracilis tendon graft was passed through the tibial and femoral drill holes, exiting on the lateral femoral condyle. The tendons were placed under approximately 15 pounds of tension while the tibial attachment was reinforced with multiple nonabsorbable sutures using figure-of-eight stitches, and the femoral attachment was secured to the lateral femoral condyle with barbed staples (Fig. 2).

The strip of iliotibial band was passed from front to back through the hole in the lateral femoral condyle and brought around the arcuate complex and under the fibular collateral ligament. It was placed under approximately 10 pounds of tension and attached to Gerdy's tubercle with a small barbed staple (Fig. 3). The defect in the iliotibial band was closed over the vastus lateralis muscle with No. 0 absorbable suture. The incisions were closed in a routine manner.

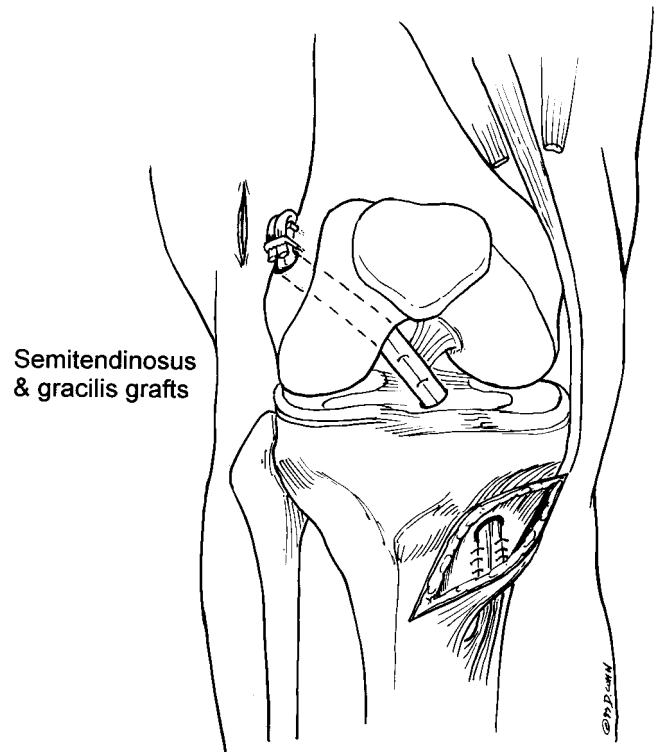


Figure 2. Patients in groups 2 and 3 had intraarticular reconstruction using the semitendinosus and gracilis tendons.

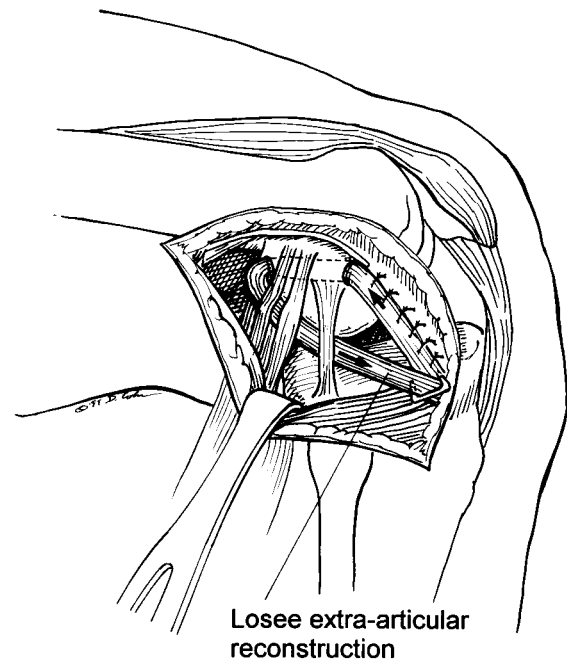


Figure 3. Patients in group 2 had an extraarticular iliotibial band tenodesis in addition to the semitendinosus and gracilis tendon intraarticular reconstruction.

Group 3—Semitendinosus and Gracilis Tendons. The patients in group 3 had ACL reconstructions with the semitendinosus and gracilis tendons. Fourteen patients in this group had a torn medial meniscus and 15 had a torn lateral meniscus. Six medial and 4 lateral menisci were repaired, and 8 medial and 11 lateral menisci were partially excised.

Reconstruction with the semitendinosus and gracilis tendons was performed with the same technique as described in group 2, although the lateral incision in group 3 was only 3 cm long.

Postoperative Regimen

The postoperative care and rehabilitation protocol were the same for all patients in the three groups. Each patient's knee was placed in a knee immobilizer postoperatively. Range of motion, quadriceps muscle sets, straight leg raises, and prone hamstring muscle stretching exercises began the day after surgery. Functional muscle stimulation was used 2 hours three times daily for 4 weeks. The patients were allowed to walk with 25% weightbearing with the aid of crutches.

At 1 week after surgery, patients were expected to have a range of motion from 0° of extension to 90° of flexion. Weightbearing was increased to 50% and active range of motion exercises were begun. At 3 weeks postoperatively, a functional brace was applied and full weightbearing was allowed. Stationary biking, active knee extension with weights, and one-quarter squats were introduced at 4 weeks after surgery.

The patients were expected to have a full and equal range of motion compared with the opposite normal knee

at 6 weeks after surgery. Stair climbing exercises with short steps and minimal resistance were started along with swimming and submaximal isokinetic exercises. After 12 weeks, the patients began a running program and high-speed, high-repetition, isokinetic exercises. Between 4 and 6 months after surgery, patients performed functional activities in the brace, including full-speed running. They were allowed to advance to full activity, including competitive sports, at 6 to 7 months after surgery.

RESULTS

One patient in group 2 and two patients in group 3 were lost to follow-up. The remaining 102 patients were reexamined at a minimum of 24 months (mean, 35.4 ± 11.6 ; range, 24 to 48) after ACL reconstruction.

There were no statistically significant differences in the postoperative range of motion among the three groups, although more patients in group 2 had limitations in motion. In group 1, three patients failed to regain between 3° and 5° of extension when compared with the contralateral knee; one patient lost 17° of flexion. In group 2, seven patients lost from 3° to 10° of extension and eight patients lost from 6° to 33° of flexion. In group 3, one patient lost 3° of extension and two lost 6° to 10° of flexion.

Each procedure was reliable in eliminating the pivot shift. Stability evaluation with the KT-1000 arthrometer (manual maximum side-to-side difference) revealed statistically significant better stability in group 1 (mean, 2.1 mm) compared with group 3 (mean, 3.1 mm) ($P < 0.05$). The incidence of patients with less than 3 mm of side-to-

TABLE 2
Postoperative Results

Category	Group ^a		
	1	2	3
Follow-up (mean $\pm$ SD; months)	34.6 $\pm$ 11.4	35.7 $\pm$ 12.1	35.9 $\pm$ 11.7
Patellofemoral crepitation (no. of patients)			
0	26	23	30
1+	8	9	3
2+	1	2	0
3+	0	0	0
Muscle strength (mean and range; % of normal side)			
Quadriceps			
60 deg/sec	86 (53–111)	83 (70–103)	96 (76–116)
180 deg/sec	91 (65–111)	87 (60–118)	99 (69–123)
Hamstring			
60 deg/sec	96 (59–117)	94 (60–125)	96 (69–123)
180 deg/sec	100 (66–147)	94 (70–131)	94 (63–133)
Objective stability (mm)			
Mean $\pm$ SD (range)	2.1 $\pm$ 2.0 (–2 to 7.3)	2.6 $\pm$ 2.2 (–2 to 9)	3.1 $\pm$ 2.3 (–0.5 to 9)
No. of patients with <3 mm side-to-side difference	25	21	17
Subjective stability			
Pivot shift (no. of patients)			
0	26	23	20
1+	2	4	5
2+	5	5	4
3+	2	2	4

^a See footnote at Table 1.

TABLE 3

Preoperative and Postoperative Hospital for Special Surgery Knee Scores for Patients with Chronic Injuries by Group (Mean $\pm$ SD)

Category (points possible)	Group ^a					
	1		2		3	
	Pre	Post	Pre	Post	Pre	Post
Subjective (25)	12.1 $\pm$ 4.2	13.1 $\pm$ 3.3	13.2 $\pm$ 4.6	15.1 $\pm$ 4.8	12.4 $\pm$ 4.1	14.0 $\pm$ 3.9
Functional (20)	5.4 $\pm$ 5.0	19.3 $\pm$ 1.2	4.1 $\pm$ 3.6	17.5 $\pm$ 3.8	4.9 $\pm$ 5.4	18.8 $\pm$ 1.8
Objective (55)	37.7 $\pm$ 5.2	46.0 $\pm$ 5.6	39.6 $\pm$ 5.1	44.3 $\pm$ 7.0	37.8 $\pm$ 6.9	48.4 $\pm$ 12.2
Total (100)	49.1 $\pm$ 14	77.4 $\pm$ 7.0	49.1 $\pm$ 12	75.8 $\pm$ 13.2	51.1 $\pm$ 10	80.0 $\pm$ 6.9

^a See footnote at Table 1.

side laxity difference was 25 of 35 (71%) in group 1, 21 of 34 (62%) in group 2, and 17 of 33 (52%) in group 3.

There were no statistically significant differences in the incidence of patellofemoral crepitation ($P < 0.2$) or the mean quadriceps and hamstring muscle strength as measured by the Cybex II dynamometer ($P < 0.06$ to 0.5) (Table 2).

There was no statistically significant difference in the Hospital for Special Surgery knee⁴ scores between groups either preoperatively or postoperatively among the patients who were treated for chronic knee injuries (Table 3). All three procedures improved mean knee scores by 27 to 30 points when evaluated with the criteria of this rating scale.

As ascertained by the IKDC evaluation for level of activity, all patients were participating in level I activities before their injury. At final follow-up, most patients in each group returned to their preinjury activity level (Table 4). There was no statistically significant difference in the IKDC subjective assessment or symptoms among the three groups (Table 5). There was a higher incidence of loss of motion and compartment crepitation in group 2 patients when compared with patients in groups 1 and 3. There was less pathologic laxity in group 1 compared with groups 2 and 3, and the difference between groups 1 and 3 was statistically significant ($P < 0.05$).

Roentgenographic evaluation by the IKDC criteria revealed no statistically significant difference among the groups. The majority of the patients were rated normal in each group. Five patients in group 1, three patients in group 2, and six patients in group 3 had mild changes. One patient in group 2 had moderate degenerative changes.

TABLE 4

Preoperative and Postoperative Activity Levels by Group^a

Activity level	Group ^b		
	1 (No.)	2 (No.)	3 (No.)
Preoperative (after injury)			
I	0	0	0
II	4	1	1
III	25	21	25
IV	0	0	0
Final follow-up			
I	29	25	29
II	5	7	2
III	1	2	2
IV	0	0	0

^a Activity level for patients with subacute or chronic injuries. I, very strenuous; II, strenuous; III, moderate; IV, sedentary.

^b See footnote at Table 1.

The final IKDC rating revealed a higher incidence of normal or near normal results in group 1 when compared with groups 2 and 3 ($P = 0.02$).

Complications

There were no infections, deep venous thromboses, nerve injuries, or other perioperative complications in this series. Patients in all three groups required additional surgery after the ACL reconstruction procedure. In group 1, one patient had arthroscopic plica removal and staple removal and another patient had an ACL reconstruction with an allograft. In group 2, six patients required additional procedures. Two patients required a manipulation

TABLE 5

International Knee Documentation Committee Knee Evaluation Results (Number of Patients with Each Rating)

Category	Group ^a		
	1	2	3
Subjective assessment			
Normal	17	15	17
Nearly normal	18	17	13
Abnormal	0	2	3
Severely abnormal	0	0	0
Symptoms			
Normal	26	23	23
Nearly normal	8	8	9
Abnormal	1	1	1
Severely abnormal	0	2	0
Range of motion			
Normal	33	23	29
Nearly normal	2	11	4
Abnormal	0	0	0
Severely abnormal	0	0	0
Ligament examination			
Normal	24	17	16
Nearly normal	11	11	10
Abnormal	0	6	7
Severely abnormal	0	0	0
Compartment findings			
Normal	32	26	31
Nearly normal	3	8	2
Abnormal	0	0	0
Severely abnormal	0	0	0
Final IKDC Rating			
Normal	11	6	8
Nearly normal	23	17	18
Abnormal	1	9	7
Severely abnormal	0	2	0

^a See footnote at Table 1.

for lack of flexion, one had an arthroscopic medial meniscectomy for failed meniscal repair, and three patients had removal of staples and arthroscopic debridement. In group 3, five patients had additional procedures. Three patients required removal of the lateral staples, one had removal of a staple and a lateral meniscectomy, and one had an arthroscopic repair of the medial meniscus and a subsequent ACL reconstruction using the middle third of the patellar tendon. The two cases that required an additional reconstruction were considered failures, and the results obtained before the subsequent ACL reconstruction were used for the analysis.

DISCUSSION

The results of our study substantiate the similarities found in previous reports that document that good or excellent results may be obtained in the majority of ACL reconstructions when using either patellar tendon or semitendinosus and gracilis tendon autografts. The goals of ACL reconstruction are to decrease symptoms, improve function, and return patients to their preinjury activity level. Patients in group 1 (patellar tendon) and group 3 (semitendinosus and gracilis reconstruction) demonstrated no statistically significant differences in these parameters. More than 80% of patients in both groups returned to their preinjury activity level, and 90% subjectively rated their knees normal or nearly normal. Comparison of preoperative and postoperative Hospital for Special Surgery knee scores in patients with chronic injuries also demonstrated statistically significant improvements in function and total score for both groups.

Harvest site abnormalities are a commonly reported problem after ACL reconstruction. Extensor mechanism pain, patellofemoral crepitation, and loss of extension have been associated with autologous patellar tendon reconstruction.^{1, 21, 32} Marder et al.²¹ reported a 24% incidence (17 of 72) of anterior knee pain in patients with either patellar tendon or doubled semitendinosus and gracilis tendon reconstructions. In contrast, our results are similar to those of Aglietti et al.² and Otero and Hutcheson,²⁵ who found that anterior knee pain was only a minor problem. Although the incidence of patellofemoral crepitation was greater in the group with patellar tendon reconstructions compared with the group with semitendinosus and gracilis tendon reconstructions, the incidence did not reach statistical significance.

Historically, limitation of extension has been a significant clinical problem after ACL reconstruction when using the central third of the patellar tendon. Early mobilization postoperatively with emphasis on full passive extension has minimized extension loss. Our results, like those of others, demonstrate no statistically significant difference in range of motion after patellar tendon reconstruction compared with semitendinosus and gracilis tendon reconstructions. This study also confirms that harvesting the semitendinosus and gracilis tendons or the middle third of the patellar tendon does not cause a permanent loss of strength in either the hamstring or quadriceps muscles.

Although our results and those of the literature suggest that, in many respects, these procedures are equivalent,

critical analysis of the data shows a clinically important difference with regard to stability. Patients who had ACL reconstruction with an autologous patellar tendon had less residual anterior translation when compared with patients who received semitendinosus and gracilis tendon autografts. Otero and Hutcheson²⁵ also found consistently greater knee stability with patellar tendon reconstruction. Aglietti et al.² and Holmes et al.¹³ demonstrated a trend toward better objective stability when using the patellar tendon as an autograft. In contrast, Marder et al.²¹ did not find a difference in residual laxity between the two graft sources. The results of our study cannot be compared directly with those of the latter studies,^{2, 21, 26} primarily because they used doubled semitendinosus and gracilis tendon grafts. Even so, residual laxity in this series, as measured by the KT-1000 arthrometer, was within the range of values reported by Marder et al.²¹ and Aglietti et al.² Although reconstruction with the semitendinosus and gracilis tendons often yields good or excellent results, there appears to be a definite trend toward better stability with autogenous patellar tendon reconstructions.

The overall knee rating, according to the IKDC scale, showed that 34 of 35 patients (97%) in group 1, 23 of 34 patients (68%) in group 2, and 26 of 33 patients (79%) in group 3 had normal or nearly normal scores. The results of group 1 were statistically significantly better than those of group 2 ($P < 0.02$). However, the IKDC method of evaluation uses multiple categories and tests. Consequently, care must be taken in claiming statistical significance of these findings.

The results of our study also show that an extraarticular procedure did not improve the results of intraarticular semitendinosus and gracilis tendon graft reconstruction. Patients in group 2 had a higher incidence of patellofemoral crepitation and loss of motion than did patients in group 3. Consequently, we no longer recommend the extraarticular procedure for routine ACL reconstructions.

In summary, ACL reconstruction with a semitendinosus and gracilis tendon autograft or a patellar tendon autograft yields similar patient-reported outcomes, although the patellar tendon autograft may provide better objective stability at a minimum follow-up of 2 years. In addition, there appears to be no benefit to combining an intraarticular ACL reconstruction with an extraarticular iliotibial band tenodesis.

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